Highlights

- Sparse, axis-aligned feature selection, not deep capacity, drives spatial transfer.
- A sparse-attention encoder (L-TAE-S) ties TabNet for best transfer, trains faster.
- Field-level aggregation is a second, substitutable lever for robust transfer.
- Model rankings invert between in-region and spatially disjoint holdout tests.
- Crop models for new farmland must be evaluated across regions, not within one.